
Backend implementation report

The backend is implemented and tested for development use, including a live inference request through NEAR's GPU TEE service. Production deployment remains subject to the infrastructure and integration work below.

Implemented

- **Encrypted inference and receipts.** Session handling, encrypted requests and responses, signed EIP-712 receipts, encrypted transcripts, persistent storage, receipt anchoring and controlled auditor exports.
 - **NEAR GPU integration.** Intel TDX and NVIDIA attestation verification, fresh nonces, approved VM profiles, binding to the actual TLS key, and verification of the provider's request/response signature.
 - **Payments.** EIP-3009 authorizations, x402 v2 for standard EVM wallets, atomic spending limits and fee accounting, and recovery after interrupted transactions. Payment and inference retries are checked for duplicate charges and duplicate execution.
 - **Models and economics.** Registration, approval, revocation, staking, rewards, provider fees and a configurable buyback interface. Versioned software TCB policies have immutable registry bindings and controlled activation. Old sessions and in-flight results cannot bypass activation.
 - **Agent runtime and SDK.** Bounded jobs with encrypted state, budgets, cancellation, an authenticated action journal and restart recovery. Claude Agent SDK tools cover gateway operations. Payment signatures and credentials stay outside model context; uncertain calls are not repeated automatically.
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Live verification

On 19 September, one request to **GLM-5.3-Flash on NEAR** completed with valid CPU attestation, evidence for eight GPUs, an approved VM profile, verified TLS binding and a valid provider signature. The encrypted response and evidence were persisted; replay after a gateway restart returned the same result without a second model call.

The same scenario verified auditor export, test-USDC settlement and receipt anchoring on a local EVM chain. It completed in **14.4 seconds** with a 32-token generation limit. This was a functional test; it did not measure TEE overhead or verify live Arc payments.

Test results

Component	Tests passed	Line coverage	Branch coverage
Core	438	99.80%	98.38%
API, including integration tests	797	97.00%	91.54%
Worker and transaction signer	160	99.78%	98.30%
Claude Agent SDK adapter	70	100%	96.93%
Solidity contracts	131	100%	91.09%
Intel/NVIDIA verifier	38	-	-
Modal and supporting scripts	61	-	-

Coverage applies to configured source files. The API run includes restart, payment-recovery, TCB rotation and concurrent agent-job tests. The unchanged verifier and helper suites were last checked on 19 September; their coverage was not measured.

Type checking passed across all TypeScript packages. The dependency audit completed on 20 September with **zero known vulnerabilities reported**. Integration tests exercised PostgreSQL, Redis and an EVM chain; temporary test containers were removed.

Remaining before production

- **Enclave-owned TEE deployment.** Move the gateway, receipt signer and agent keys into an approved confidential VM. NEAR remains the GPU provider; its inference API key does not provision a VM for our code. A switch to Phala is not required.
- **Workload assurance.** Verify the deployed application and model weights against reproducible measurements. The current development policy retains trust in NEAR's deployment manager. The Solidity registry does not independently verify vendor hardware evidence.
- **Network and payment integration.** Deploy matching contracts on the target network, validate real USDC transfers, integrate Arc's confidential-transfer interface and configure a live DEX adapter. Public token transfers remain visible on-chain.
- **Agent operations and performance.** Hardware wallet custody and external-tool execution remain open. Benchmark comparable GPU workloads with and without confidential computing. Collateralized credit and slashing require agreed product rules.

The current NEAR development policy expires on 26 September 2026 at 16:48 UTC and requires review before renewal. Production mode remains disabled while these deployment requirements are open.

Scope: Enclave Backend and Enclave Overview specifications. Core, API, worker, SDK and contract checks updated 20 September 2026; dependency audit completed the same day.